WEEK 12 · Continuous time

Continuous-Time Financial Models

From Brownian motion to mean reversion, short rates and bond prices

Brownian motion · SDE · GBM · Itô · OU · AR(1) · Vasicek · CIR · bond prices · Monte Carlo

Brownian motion supplies the random shocks; an SDE attaches a rule of change to them.
OU is a rule that returns to the mean; Vasicek applies it to the short rate.

Continuous-time models keep shocks and rules separate

Three messages that run through this lecture

1

Shocks grow with the square root of time

The variance of a Brownian increment is proportional to time, so squared terms survive in stochastic calculus.

2

The exact discretization of OU is an AR(1)

Mean-reversion speed, long-run mean and shock size can be recovered from daily or monthly regressions.

3

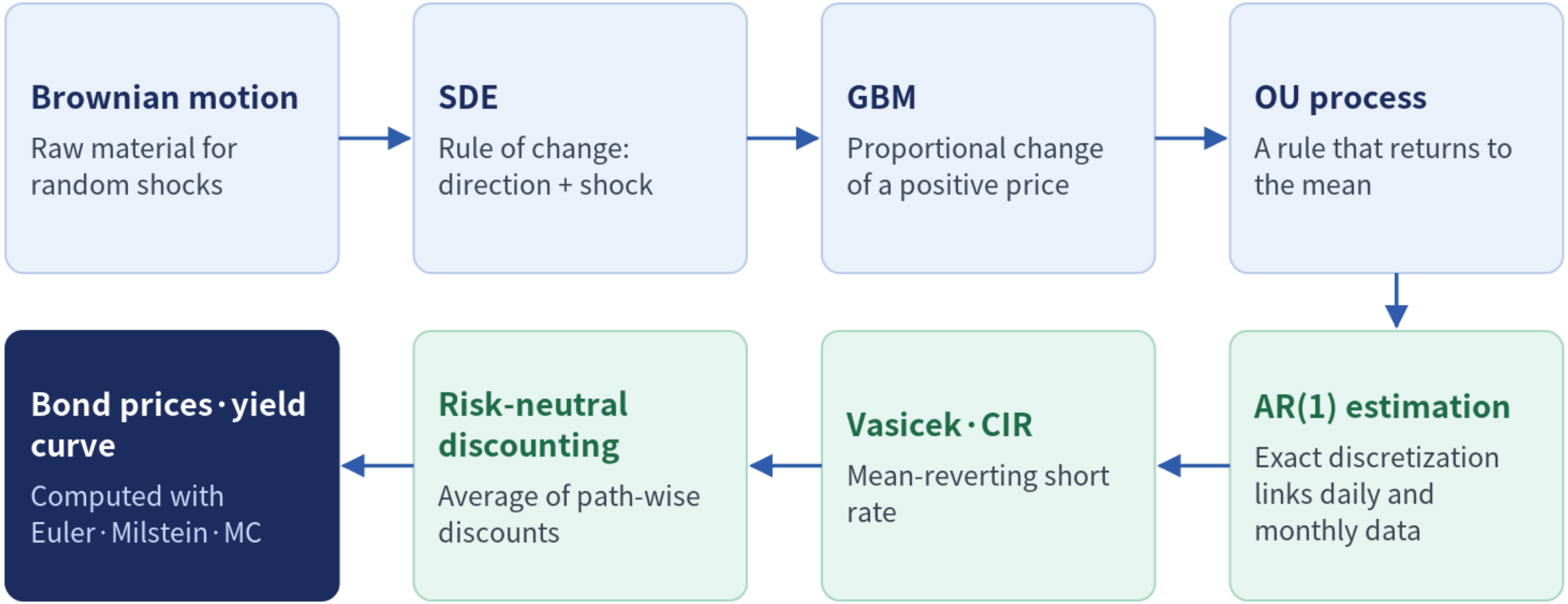
Bond prices: discount path by path, then average

The price is the expected discount factor under the risk-neutral measure. Keep forecasting and pricing probabilities apart.

Continuous time is a useful mathematical approximation, not a replica of reality — tick sizes, asynchronous trading and jumps remain

One line runs from random shocks to bond prices

Each step uses the tools of the previous one



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PART 1 · Random change

Brownian motion · SDE · GBM · Itô

Shock size · rule of change · price model · Itô correction

The variance of a Brownian increment is proportional to time

Standard Brownian motion: starts at 0, continuous paths, independent changes over non-overlapping intervals

$$W_{t+\Delta} - W_t \sim \mathcal{N}(0, \Delta)$$

$$\Delta W = \sqrt{\Delta}\, \underbrace{Z}_{\text{standard normal draw}}$$

Variance = time; std. dev. = $\sqrt{\ }$ time

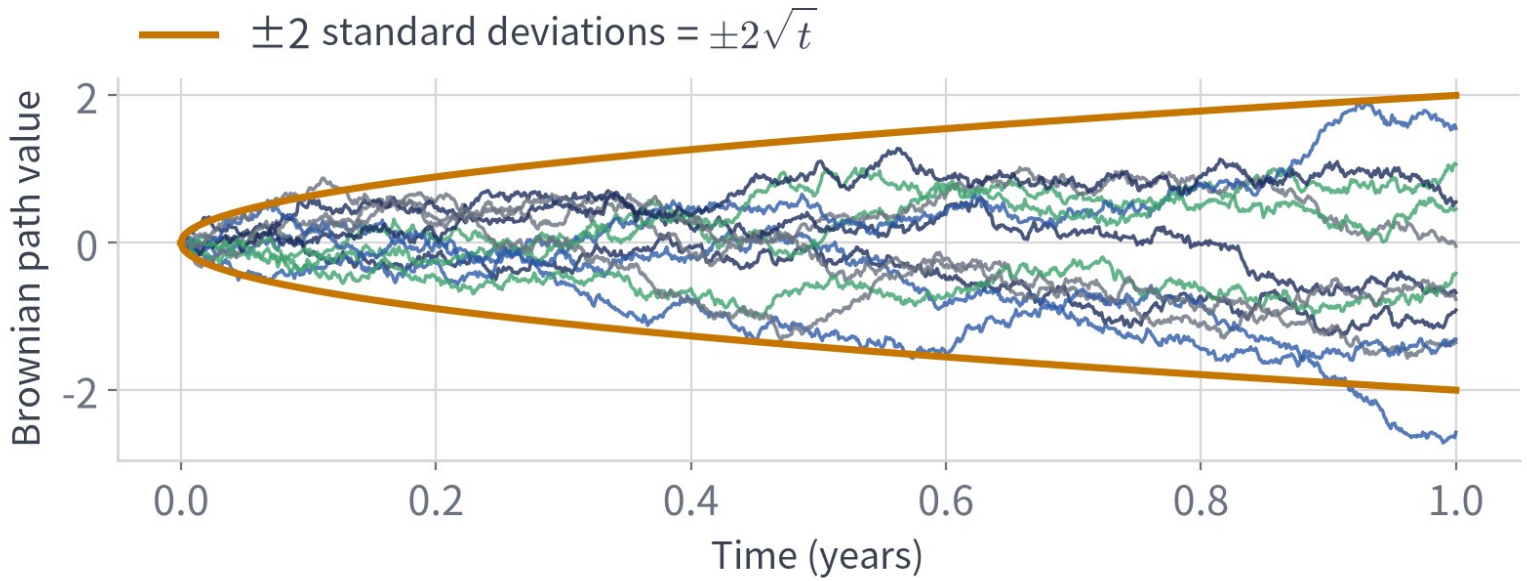
- Interval 1 → std. dev. 1
- Interval 1/4 → std. dev. 1/2
- Interval 1/100 → std. dev. 1/10

Why it matters

- Cut time to 1/100 and the shock is still 1/10
- So squared terms cannot be ignored in stochastic calculus

Brownian paths are continuous but not smooth

The mean change is 0, but the spread of possible values widens over time · 12 paths · ※ synthetic paths



Nowhere differentiable

No jumps, yet not differentiable in the usual sense at any point

What mean 0 means

It does not mean each path stays near 0

An SDE combines direction and shock in one equation

The two terms of a stochastic differential equation (SDE)

$$dX_t = \underbrace{a(t, X_t)\, dt}_{\text{drift: conditional mean change}} + \underbrace{b(t, X_t)\, dW_t}_{\text{diffusion: shock size}}$$

$$X_{t+\Delta} - X_t \approx a(t, X_t)\Delta + b(t, X_t)\sqrt{\Delta} Z$$

$$E[\Delta X \mid X_t] \approx a(t, X_t)\Delta, \qquad \mathrm{Var}(\Delta X \mid X_t) \approx b(t, X_t)^2 \Delta$$

How a computer reads it

Over a short step, the mean change is the drift and the variance is proportional to the squared diffusion

Misconception

If the state is random, the drift is random along the whole path too

Match the time unit of the coefficients to the step size

An annual model computed by trading day usually takes a step of 1/252

$$\underbrace{0.20}_{\text{annual volatility}} \sqrt{1/252} \approx 0.01260 = 1.260\%$$

Choosing the step

- Trading day 1/252 · monthly 1/12
- For rates, the day-count convention can change this

Common mistake

- Annual coefficients with a one-day step of 1 mismatch the units

In GBM it is the log return that is normal

Geometric Brownian motion — the basic model in which prices move in proportion

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

$$S_{t+\Delta} = S_t \exp\left[(\mu - \tfrac{1}{2}\sigma^2)\Delta + \sigma\sqrt{\Delta}Z\right]$$

$$\underbrace{\log \frac{S_{t+\Delta}}{S_t}}_{\text{log return}} \sim \mathcal{N}\left((\mu - \tfrac{1}{2}\sigma^2)\Delta, \sigma^2\Delta\right)$$

Properties

- A positive initial value keeps the exact solution positive
- The price is lognormal

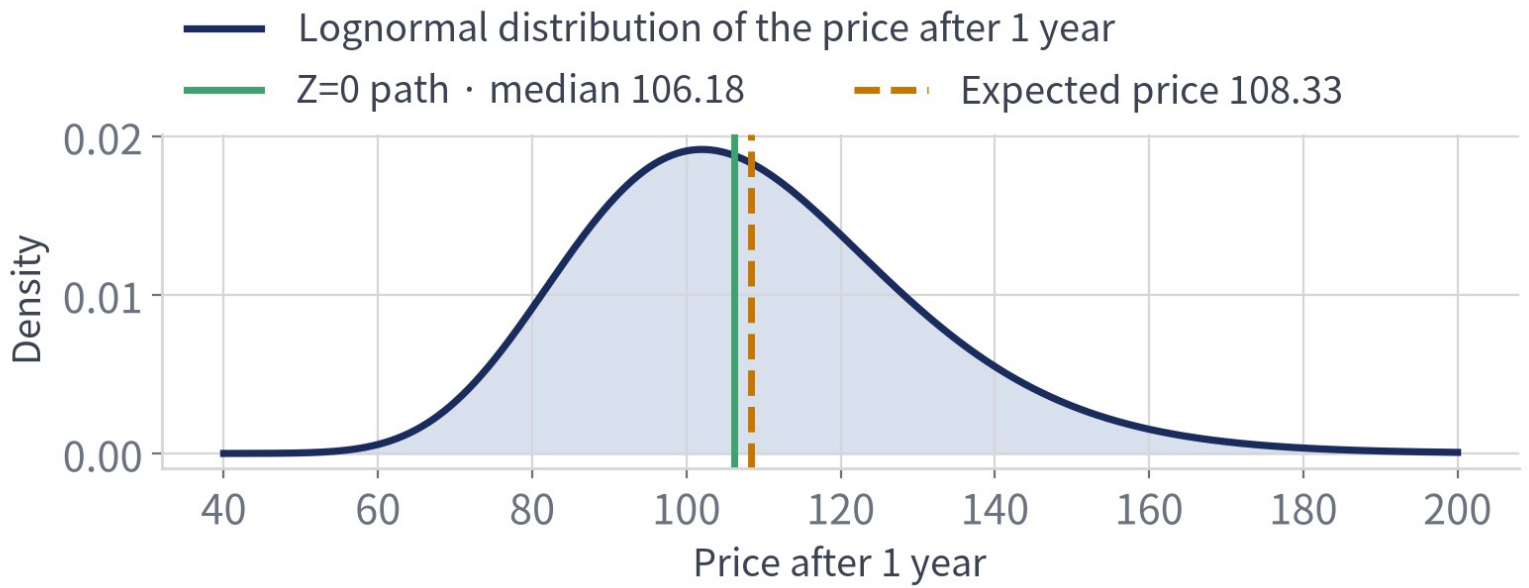
Limits

- Does not explain volatility clustering or jumps by itself
- The simple return is a lognormal minus 1

The zero-shock path is the median, not the expected price

Start price 100 · expected return 8% · volatility 20% · 1 year

$$\underbrace{S_1 = 100e^{0.08-0.5(0.20)^2} = 100e^{0.06} \approx 106.18}_{Z=0 \text{ path: conditional median}} \qquad \underbrace{E[S_1] = 100e^{0.08} \approx 108.33}_{\text{expected price}}$$



How to read it

- Z=0 path: 106.18
- Expected price: 108.33
- Gap = skewness of the lognormal

Changes passed through a function carry a second-order correction

Itô's lemma — the ordinary chain rule plus a second-derivative term

$$df = \left(f_t + a f_x + \underbrace{\frac{1}{2} b^2 f_{xx}}_{\text{Itô correction}} \right) dt + b f_x dW_t$$

$$(dW_t)^2 = dt, \qquad dt\, dW_t = 0, \qquad (dt)^2 = 0$$

Why the squared term survives

A Brownian increment is of size $\sqrt{\text{time}}$ → squared, it is of size time

Precise meaning

It is the rule that quadratic variation converges to time

Applying Itô to the log price yields the correction term

Advanced — plug in the first and second derivatives of log to derive the exact GBM solution

$$f_S = \frac{1}{S}, \qquad f_{SS} = -\frac{1}{S^2}$$

$$d\log S_t = \frac{1}{S_t}(\mu S_t dt + \sigma S_t dW_t) + \frac{1}{2}\left(-\frac{1}{S_t^2}\right)\sigma^2 S_t^2 dt$$

$$\boxed{d\log S_t = (\mu - \tfrac{1}{2}\sigma^2)dt + \sigma dW_t}$$

Result

- The log-price drift is the expected return minus half the variance

Link

- Integrate over time and exponentiate: the exact GBM solution

02

PART 2 · Mean reversion

The OU process and AR(1) estimation

Conditional distribution · half-life · exact discretization · pair spreads

The farther from the mean, the harder OU pulls back

Ornstein–Uhlenbeck process — a drift that pulls toward the long-run mean like a spring

$$dX_t = \underbrace{\kappa}_{\text{reversion speed}}\left(\underbrace{\theta}_{\text{long-run mean}} - X_t\right)dt + \underbrace{\sigma}_{\text{shock size}}\, dW_t, \qquad \kappa > 0$$

When a parameter rises

- Long-run mean: the level it returns to rises
- Reversion speed: the expected gap shrinks faster
- Shock size: moves over the same period get larger

What ‘reversion’ means

- It is a property of the conditional mean
- Above the mean, it can still rise tomorrow — the shock can overpower the pull

The expected gap decays exponentially; the variance saturates

Conditional mean and variance given the current state, and the long-run stationary distribution

$$\boxed{E[X_{t+h} \mid X_t] = \theta + (X_t - \theta)e^{-\kappa h}}$$

$$\boxed{\mathrm{Var}(X_{t+h} \mid X_t) = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa h})}$$

$$\boxed{\underbrace{X_\infty}_{\text{stationary distribution}} \sim \mathcal{N}\left(\theta, \frac{\sigma^2}{2\kappa}\right)}, \qquad s_\infty = \frac{\sigma}{\sqrt{2\kappa}}$$

$$e^{-\kappa h_{1/2}} = \tfrac{1}{2} \implies \boxed{\underbrace{h_{1/2}}_{\text{half-life}} = \frac{\log 2}{\kappa}}$$

Keep apart

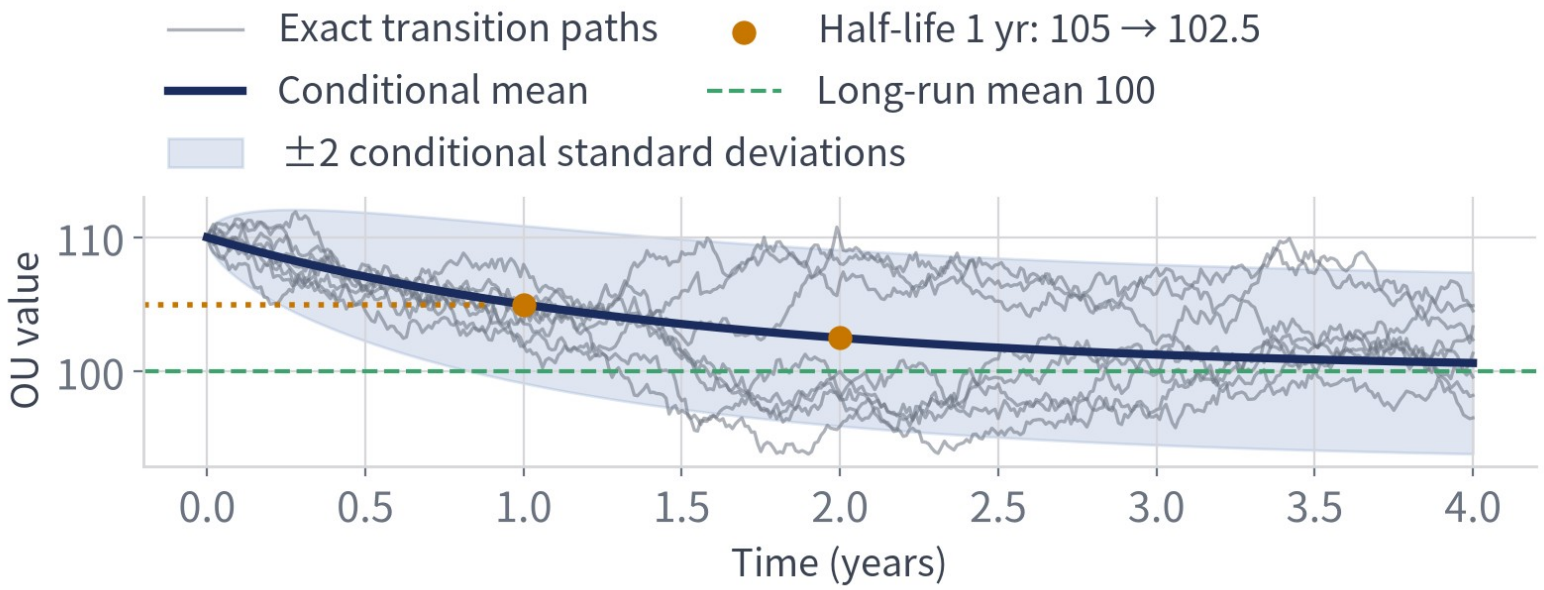
Shock coefficient ≠ std. dev. of the long-run distribution

Half-life

Expected-gap halving time — not arrival time

With a 1-year half-life, the expected gap halves every year

Now 110 · long-run mean 100 · reversion speed log2/yr · shock 4 · ※ synthetic paths



Expected value

105 after 1 year, 102.5 after 2 · std. dev. after 1 year ≈ 2.94

Paths have a distribution

The value after 1 year is not bound to be 105

OU observed at equal intervals is exactly an AR(1)

The persistence and constant of the exact transition map to the AR(1) coefficients

$$\boxed{X_{t+\Delta} = \theta + (X_t - \theta)e^{-\kappa\Delta} + \epsilon_t}, \quad \epsilon_t \sim \mathcal{N}\left(0, \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta})\right)$$

$$X_{n+1} = c + \phi X_n + \epsilon_n, \qquad \boxed{\phi = e^{-\kappa\Delta}}, \qquad \boxed{c = \theta(1 - \phi)}$$

Conditions

- Equally spaced observations
- Independent shocks across intervals

Meaning

- A continuous-time model can be estimated by daily or monthly regression

Map regression results back to continuous-time coefficients

Use the residual variance to recover reversion speed, long-run mean, shock size and half-life

$$\kappa = -\frac{\log \phi}{\Delta}, \qquad \theta = \frac{c}{1-\phi}, \qquad \sigma = \sqrt{\frac{2\kappa s_\epsilon^2}{1-\phi^2}}, \qquad h_{1/2} = -\frac{\Delta \log 2}{\log \phi}$$

When it applies

- The exact mapping needs persistence between 0 and 1
- At or above 1, or at or below 0, it cannot be this OU's transition coefficient

Common mistake

- Do not set the shock size equal to the residual std. dev.
- Do not plug any regression output into the formulas

Monthly persistence 0.9 is a half-life of about 6.6 months

Monthly regression: constant 0.2 · persistence 0.9 · residual std. dev. 0.5 · step 1/12 yr

$$\kappa = -12\log(0.9) \approx 1.2643 \ / \,\mathrm{yr}, \qquad \theta = 0.2/(1-0.9) = 2$$

$$\sigma = \sqrt{\frac{2(1.2643)(0.5^2)}{1-0.9^2}} \approx 1.8241, \qquad h_{1/2} \approx 0.5482\,\mathrm{yr} \approx 6.5788\,\mathrm{months}$$

When it becomes unstable

- With short samples or persistence very close to 1, the half-life estimate wobbles

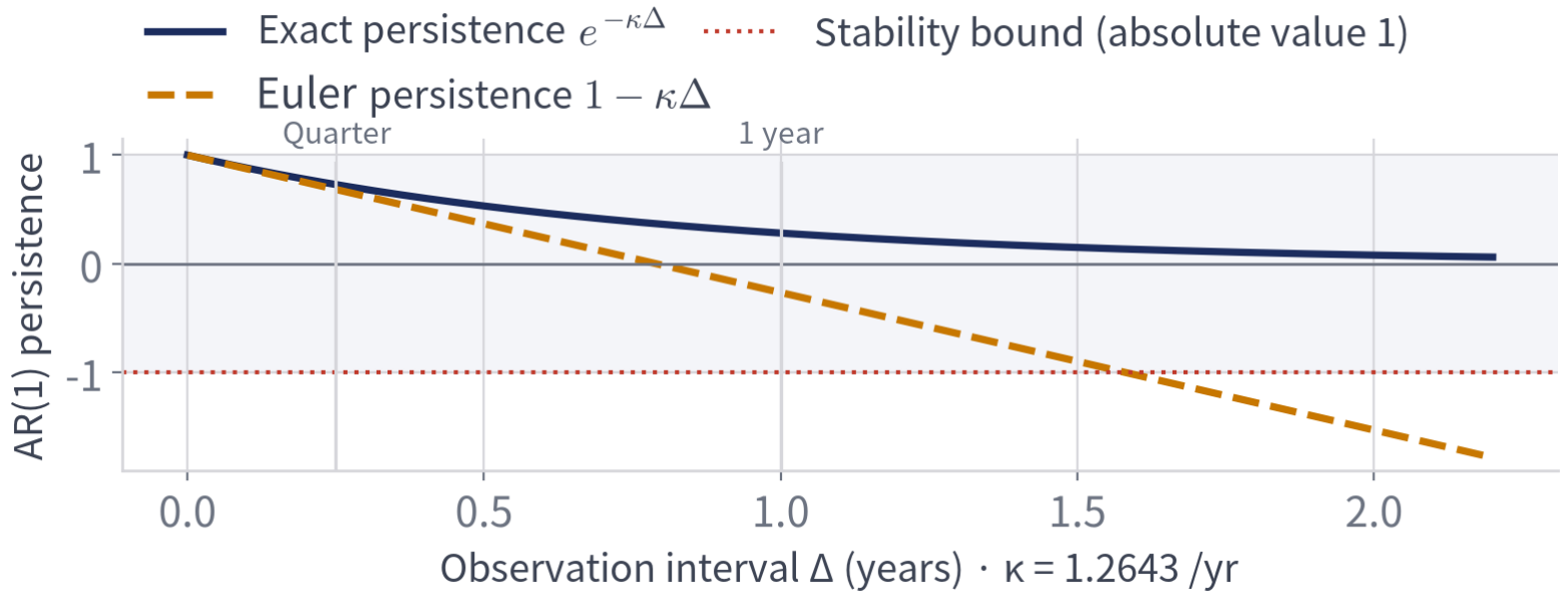
Estimation

- Under the likelihood conditional on the first observation, AR(1) regression is convenient
- An initial distribution or constraints can change the result

With wide observation intervals, the Euler approximation breaks down

Exact and Euler persistence agree only for small steps

$$\underbrace{X_{t+\Delta} \approx X_t + \kappa(\theta - X_t)\Delta + \sigma\sqrt{\Delta}Z}_{\text{Euler approximation of OU}}$$



Differences

- The shock variance also differs
- Large steps can be non-stationary or oscillate
- Use the exact formula on grid points

OU measures spread deviations in long-run std. devs.

When a two-asset spread mean-reverts stably, OU is a candidate model

$$X_t = P_t^A - \beta P_t^B \qquad z_t = \frac{X_t - \hat{\theta}}{\hat{\sigma}/\sqrt{2\hat{\kappa}}}$$

z_t: deviation in long-run standard deviations

What OU does not give

- An entry threshold of 2 or profitability does not follow from OU
- A half-life is no guaranteed exit date or loss limit

Validation order

- Estimate on the past only → trade the later period
- Include costs, slippage and borrow costs → parameter stability

Cointegration is not sufficient for OU — nonlinearity, heteroskedasticity and jumps may remain

03

PART 3 · Rates and bonds

Short rates and bond prices

Vasicek · CIR · physical vs risk-neutral probability · affine prices

Vasicek is OU with the short rate as the state

Instantaneous short rate: growth rate of a risk-free account over a very short period

$$d \underbrace{r_t}_{\text{instantaneous short rate}} = \kappa(\theta - r_t)dt + \sigma dW_t$$

What carries over

- OU conditional mean and variance, exact discretization, half-life
- Normal distribution, so rates can turn negative

Rate units

- 3% is 0.03, 1%p is 0.01, 1bp is 0.0001
- A shock coefficient of 0.01 is a shock of about 1%p a year

Proxying with the policy rate or 3-month rate can add error

CIR scales the shock size with the rate level

Same drift as Vasicek; diffusion proportional to the square root of the rate

$$dr_t = \kappa(\theta - r_t)dt + \underbrace{\sigma\sqrt{r_t}}_{\text{level-dependent diffusion}} dW_t$$

$$\underbrace{2\kappa\theta \geq \sigma^2}_{\text{Feller condition}}$$

Versus Vasicek

- Transition: normal vs scaled non-central chi-square
- Negative: possible vs non-negative
- The two models' shock coefficients have different units

Feller condition

- If it holds and the start is positive, 0 is never reached
- If it fails, 0 can be reached, but never negative values

CIR is sampled exactly from a non-central chi-square

Advanced — rather than memorizing the formula, remember that grid-point rates can be generated exactly and non-negative

$$r_{t+\Delta} = c\,Z, \qquad Z \sim \chi'^2_d(\lambda)$$

$$c = \frac{\sigma^2(1-e^{-\kappa\Delta})}{4\kappa}, \qquad d = \frac{4\kappa\theta}{\sigma^2}, \qquad \lambda = \frac{4\kappa e^{-\kappa\Delta} r_t}{\sigma^2(1-e^{-\kappa\Delta})}$$

Use

- A grid-point generator that avoids Euler's negative values

Keep forecasting and pricing probabilities apart

Physical probability P comes from historical rate series; risk-neutral Q from bond and derivative prices

$$\underbrace{dr_t = \kappa_P(\theta_P - r_t)dt + \sigma dW_t^P}_{\text{physical probability: forecasting}} \qquad \underbrace{dr_t = \kappa_Q(\theta_Q - r_t)dt + \sigma dW_t^Q}_{\text{risk-neutral probability: pricing}}$$

What risk-neutral means

- It does not mean investors ignore risk
- Risk compensation enters via a change of probability and drift

Caution

- The drift coefficients of the two equations generally differ
- To price with historical AR(1) estimates, state the risk-premium assumption

The bond price is the risk-neutral expected discount factor

Risk-free zero-coupon bond paying 1 at maturity · single-discount model, no credit or liquidity adjustment

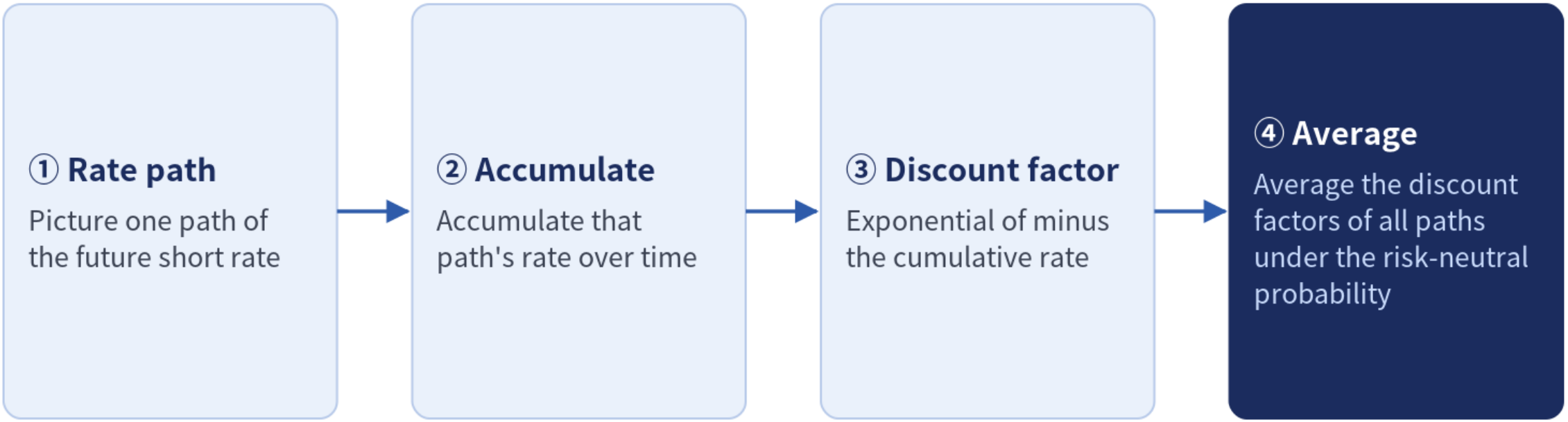
$$\underbrace{P(t,T)}_{\text{zero-coupon bond price}} = E^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_s ds\right)\middle|\mathcal{F}_t\right]$$

Steps

- One path → accumulate rates → exponential discount → average over paths
- With a constant rate this is familiar continuous-compounding discounting

Discount each path, then average

What is discounted is the future cash flow, not the future rate itself



Discounting once at the average rate gives a different price

By hand: two paths with risk-neutral probability 1/2 each, 1-year cumulative rates 0 and 0.10

$$E^{\mathbb{Q}}[e^{-I}] \neq e^{-E^{\mathbb{Q}}[I]}, \qquad I = \int_t^T r_s ds$$

$$P = \tfrac{1}{2}e^{0} + \tfrac{1}{2}e^{-0.10} \approx 0.952419 \qquad \text{vs} \qquad e^{-0.05} \approx 0.951229$$

Result

- Average of path discounts 0.952419
- Discount at average cumulative rate 0.951229

Why

- Convexity of the exponential — a teaching example of the discounting logic

Turn prices into continuously compounded zero yields

Yields at every maturity form the zero-coupon yield curve

$$\underbrace{y(t,T)}_{\text{zero-coupon yield}} = -\frac{\log P(t,T)}{\tau}$$

$$P_{\text{coupon}}(t) = \sum_i C_i P(t,T_i) + NP(t,T)$$

Distinguish

- Zero yields differ from coupon-bond yields to maturity and par yields

Coupon bonds

- Discount each cash flow at its maturity's zero price and add

The Vasicek bond price has exponential-affine form

Closed form of risk-neutral Vasicek — all coefficients are risk-neutral model coefficients

$$P(t,T) = A(\tau)e^{-B(\tau)r_t}, \qquad B(\tau) = \frac{1-e^{-\kappa\tau}}{\kappa}$$

$$\log A(\tau) = \left(\theta - \frac{\sigma^2}{2\kappa^2}\right)(B(\tau)-\tau) - \frac{\sigma^2}{4\kappa}B(\tau)^2$$

What affine means

- Log price, yield = ‘constant + coefficient × rate’
- The price itself is not linear

Rate sensitivity

- When the short rate rises, the price falls

Log price and yield are straight lines in the short rate

For given parameters and maturity, the short-rate sensitivity is always negative

$$\underbrace{\log P(t,T) = \log A(\tau) - B(\tau) r_t}_{\text{log price: affine function of the short rate}} \qquad y(t,T) = -\frac{\log A(\tau)}{\tau} + \frac{B(\tau)}{\tau} r_t$$

$$\frac{\partial P}{\partial r_t} = -B(\tau) P < 0$$

Meaning

- One current short rate moves the whole curve
- The model coefficients must also be set

By hand

The 1-year Vasicek price is about 0.9684, a yield of 3.21%

Risk-neutral reversion speed 0.5 · long-run level 4% · shock 0.01 · current short rate 3% · 1-year maturity

$$B(1) = \frac{1 - e^{-0.5}}{0.5} \approx 0.786939, \qquad \log A(1) \approx -0.00851081$$

$$\log P \approx -0.00851081 - (0.786939)(0.03) = -0.03211896$$

$$\boxed{P \approx 0.968391}, \qquad \boxed{y \approx 3.21190\%}$$

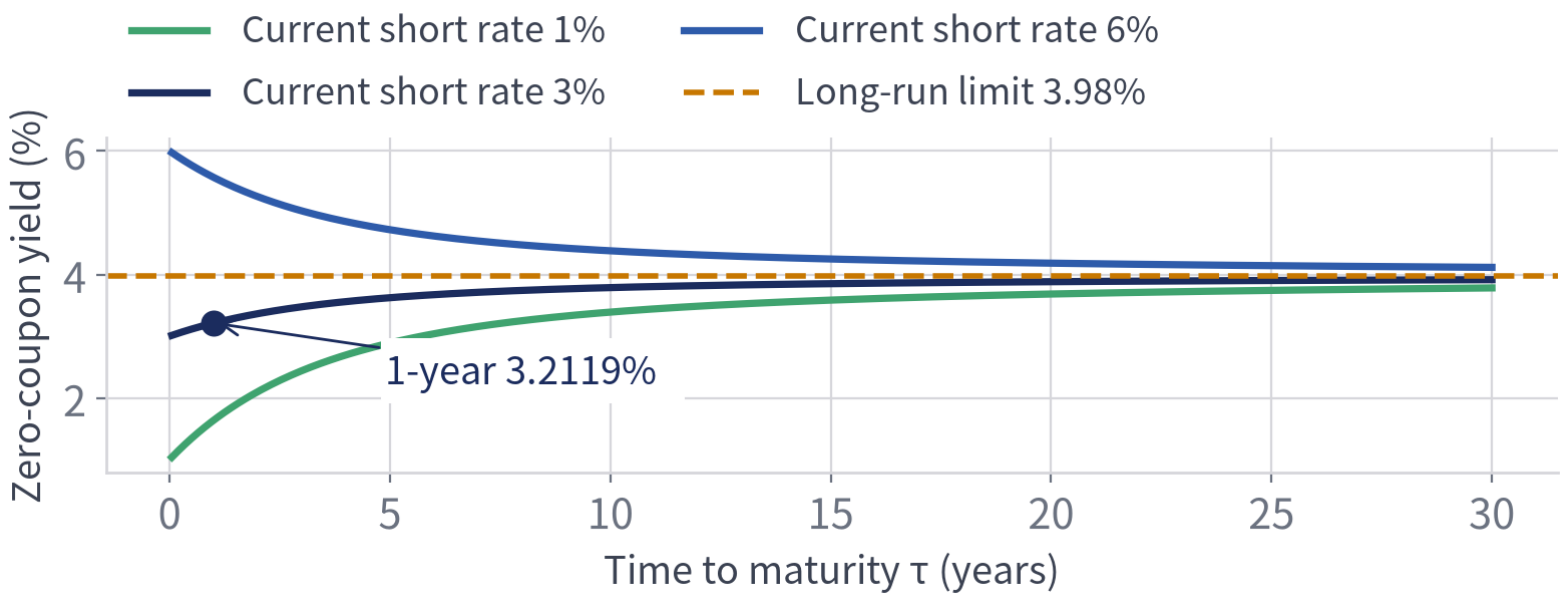
How to read it

- With face value 100 the price is about 96.8391
- The long-run level is 4%, so the 1-year yield exceeds 3%

The current rate and the coefficients set the curve shape together

Risk-neutral reversion speed 0.5 · long-run level 4% · shock 0.01 — curve computed from the formula

$$\lim_{\tau \to \infty} y(t, t+\tau) = \theta_Q - \frac{\sigma^2}{2\kappa_Q^2}$$



How to read it

- Both upward and downward slopes possible
- Even very long maturities do not equal the long-run level
- Not a simple average of future physical rates

The CIR price has the same exponential-affine structure

Advanced — before the coefficient formulas, grasp that both models are exponential-affine

$$\gamma = \sqrt{\kappa^2 + 2\sigma^2}, \qquad B(\tau) = \frac{2(e^{\gamma\tau} - 1)}{(\gamma + \kappa)(e^{\gamma\tau} - 1) + 2\gamma}$$

$$A(\tau) = \left[\frac{2\gamma e^{(\kappa+\gamma)\tau/2}}{(\gamma + \kappa)(e^{\gamma\tau} - 1) + 2\gamma}\right]^{2\kappa\theta/\sigma^2}, \qquad P = Ae^{-Br_t}$$

Summary

- In both Vasicek and CIR the price is A times an exponential discount term

04

PART 4 · Computation

Discretization and Monte Carlo

Euler · Milstein · CIR boundary · Monte Carlo bond prices · state-space link

Milstein adds one more correction for state-dependent diffusion

With enough smoothness, strong order is 1/2 for Euler and 1 for Milstein

$$\underbrace{X_{n+1} = X_n + a(X_n)\Delta + b(X_n)\sqrt{\Delta}Z_n}_{\text{Euler–Maruyama}}$$

$$\underbrace{X_{n+1} = X_n + a(X_n)\Delta + b(X_n)\sqrt{\Delta}Z_n + \frac{1}{2}b(X_n)b'(X_n)\Delta(Z_n^2 - 1)}_{\text{Milstein: state-dependent diffusion correction}}$$

Vasicek

- Constant diffusion, so the correction is 0 — Milstein = Euler

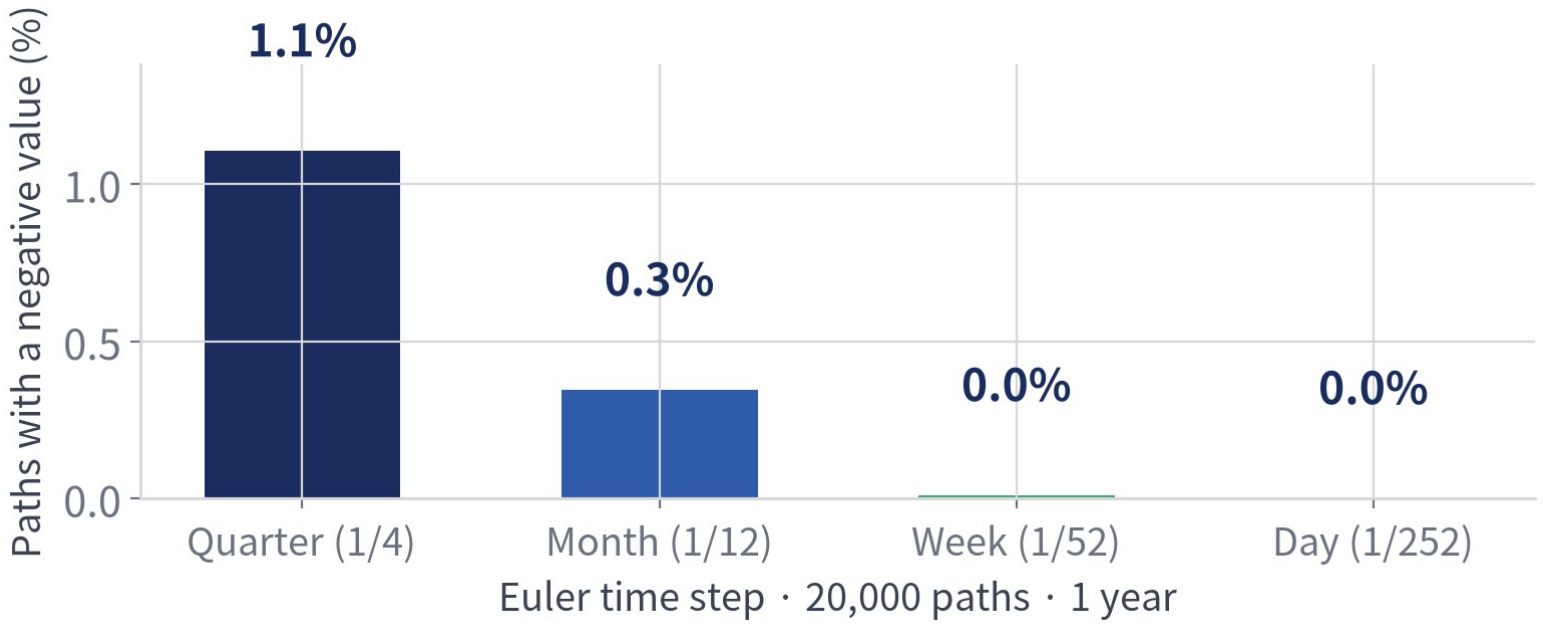
CIR

- A correction appears, but staying positive near 0 is still an issue

Euler produces negatives even when the Feller condition holds

Reversion speed 0.5 · long-run level 4% · shock 0.10 · start 0.5% · Feller 0.04 ≥ 0.01 holds

$$\underbrace{r_{n+1} = r_n + \kappa(\theta - r_n)\Delta + \sigma\sqrt{r_n}\sqrt{\Delta}Z_n}_{\text{plain Euler for CIR}}$$



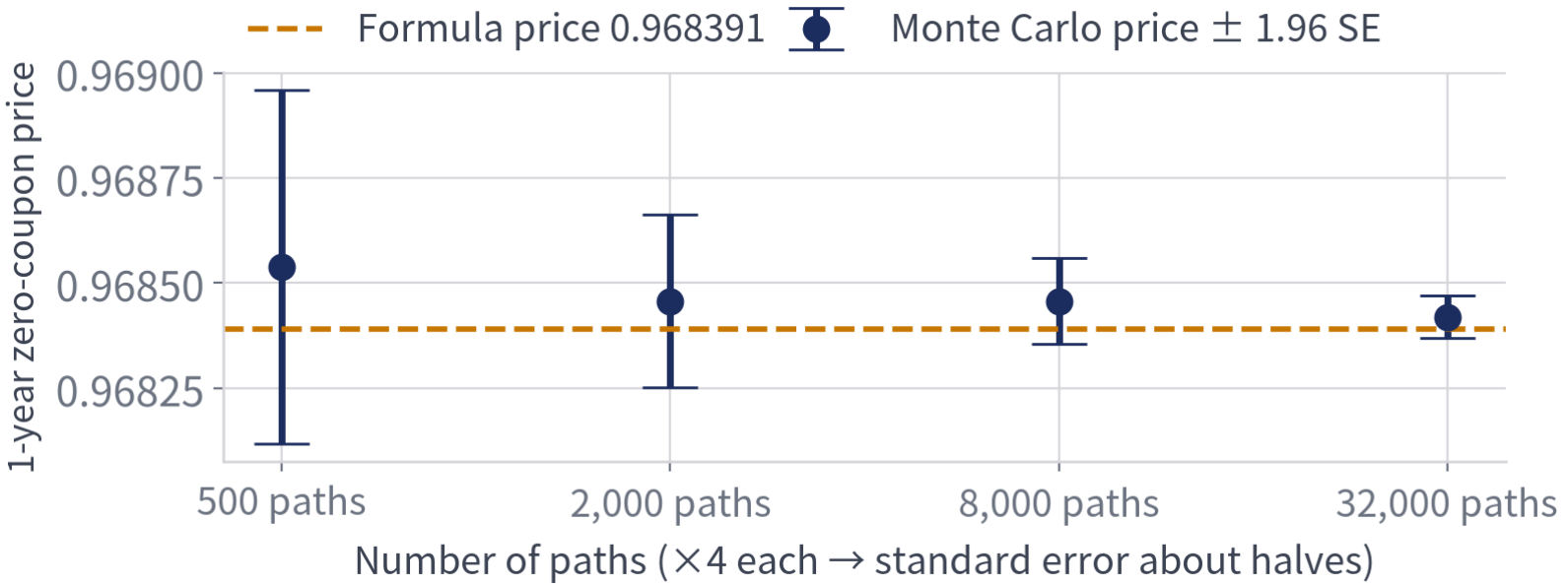
Distinguish

- Negatives are a problem of the approximation, not the model
- Use the exact transition on grid points
- If you truncate, say so

Four times the paths halves the standard error

Vasicek 1-year · exact OU transition · daily trapezoidal integration

$$D^{(m)} \approx \exp\left(-\sum_{n=0}^{L-1} r_n^{(m)} \Delta\right), \qquad \boxed{\hat{P} = \frac{1}{M}\sum_{m=1}^{M} D^{(m)}}, \qquad \boxed{\mathrm{SE}(\hat{P}) = \frac{s_D}{\sqrt{M}}}$$



Remaining error

- Discretization bias does not shrink with more paths
- Integrating rates between grid points is also an approximation
- Parameter and model error

OU plus observation error is estimated with the Kalman filter

Exact OU discretization + observation equation = linear Gaussian state-space model (link to Week 10)

$$x_{n+1} = \theta(1-\phi) + \phi x_n + w_n, \qquad \phi = e^{-\kappa\Delta}, \qquad y_n = x_n + v_n$$

$$\mathrm{Var}(w_n) = \frac{\sigma^2}{2\kappa}(1-\phi^2), \qquad \underbrace{y_n(\tau) = a(\tau) + b(\tau) r_n + v_n(\tau)}_{\text{observation equation for multi-maturity yields}}$$

Irregular spacing

- Change persistence and state-shock variance for each interval

Rate application

- Estimate the hidden short rate from yields at several maturities
- Transitions can rest on P, price-equation coefficients on Q

CIR transitions are not Gaussian, so the standard Kalman filter is not exact

Turn common wrong statements into precise ones

Continuous time · mean reversion

Statement	Precise understanding
Drawn finely enough, a Brownian path becomes smooth	Continuous but almost surely nowhere differentiable
GBM returns are normal	Log returns over a fixed interval are normal
After one half-life it reaches the mean	The expected gap halves
OU is trapped within a range	Stationary variance is finite, but there is no fixed boundary
Every stationary AR(1) is a scalar OU	The exact mapping needs persistence between 0 and 1

Rates and numerics: tell properties from approximations

Rates · prices · numerics

Statement	Precise understanding
Feller failure means negative values	Reaching 0 and going negative are different
CIR stays non-negative even with Euler	Plain Euler can produce negative values
Historical estimates give bond prices directly	Separate physical, risk-neutral probability and the risk premium
With many paths it always converges to the formula	Fixed-grid bias and integration error can remain

Confirm the key properties with questions

Answers in the chips on the right

Q1	Cut time to 1/4: shock std. dev.?	**Half**
Q2	OU above the mean: can the next value be higher?	**Yes · shock**
Q3	Reversion speed doubles: half-life? Only shock size up?	**Half · unchanged**
Q4	Does Vasicek get a Milstein correction?	**No · constant diffusion**

Lab A — compare Brownian motion, GBM and OU

No external data, seed=42, time in years throughout

[Input] Prompt · Claude Code

```
Using numpy and matplotlib, write and run an intro
lab on continuous-time models.
Use no external data, seed=42, and measure time in years.
1. Accumulate Brownian increments sqrt(dt)*Z and plot paths.
2. For GBM use the exact solution with S0=100, mu=0.08, sigma=0.20.
   Check by Monte Carlo the 1-year log-return mean 0.06 and
   variance 0.04. Distinguish simple and log returns.
3. Set OU to X0=110, theta=100, kappa=log(2), sigma=4.
   Use the exact transition; check 1-year mean 105, variance ≈ 8.6562.
4. Generate the same OU by Euler and compare errors as dt varies.
   For path comparison, sum fine-grid Brownian increments so the
   same shocks are used. When comparing individual exact and Euler
   paths, explain the correct coupling of the exact OU transition's
   weighted Brownian integral with the increments. If coupling is hard,
   compare the distribution's mean/variance errors, not strong path errors.
5. State units, dt and path count in chart titles and tables.
   Explain that the half-life is the time for the expected gap to halve,
   not an actual arrival time.
```

What to check

- Path charts
- Table of theoretical vs sample mean and variance
- Does sampling error shrink with more samples?
- Do not demand agreement to the decimal

Lab B — recover OU coefficients from AR(1)

600 monthly points · true persistence 0.9 · long-run mean 2

[Input] Prompt · Claude Code

```
Generate 600 monthly OU points with the exact transition.
theta=2, kappa=-12*log(0.9),
Use sigma=sqrt(2*kappa*0.25/(1-0.9^2)).
Draw the initial value from the stationary distribution; fix seed=42.

1. Regress X[n+1] on a constant and X[n].
2. Print c, phi, residual variance; convert to OU coefficients and half-life.
3. Compare with true phi=0.9, theta=2, half-life ≈ 6.5788 months.
4. If estimated phi is outside 0–1, do not convert; explain why.
5. Shorten the data and check over many repetitions
   whether estimation uncertainty grows.
   Do not demand that a single simulation
   match the true values exactly.
```

What to check

- Regression results and coefficient-recovery table
- Do monthly and annual half-lives agree?
- Half-life wobble in short samples

Lab C — match Vasicek prices by formula and MC

Risk-neutral coefficients · 20,000 paths · step-size sensitivity

[Input] Prompt · Claude Code

```
Write and run a Vasicek zero-coupon bond pricing lab.
Use risk-neutral kappaQ=0.5, thetaQ=0.04, sigma=0.01,
current rate r0=0.03 and a 1-year maturity.
1. Implement the Vasicek A, B formulas; compute the price and
   continuously compounded yield. Compare with ≈0.968391 and ≈3.21190%.
2. Generate risk-neutral rates with the exact OU transition.
3. Approximate each path's cumulative rate by the trapezoid rule; discount.
4. With 20,000 paths, print the price, Monte Carlo standard error
   and 95% error interval.
5. Shrink the step to 1/12, 1/52, 1/252 and compare integration errors.
   Explain that, due to sampling error, a single run's error may
   not decrease monotonically.
6. Show formula prices and zero yields for 1, 2, 5, 10 years as a table and curve.
7. State that even the exact transition is not exact for the rate integral.
8. Explain that these are given as Q coefficients, and why historical AR(1)
   estimates must not be used for pricing as they are.
```

What to check

- Formula vs MC table
- Sensitivity to path count and step size
- Yield curve
- Standard error vs bias

Lab D — see the CIR boundary and numerical error

Exact transition vs coarse monthly Euler

[Input] Prompt · Claude Code

```
Compare CIR's non-negativity as a model with its Euler approximation.
Use annual units, kappa=0.5, theta=0.04, sigma=0.10 and
r0=0.005.
1. Check the Feller condition numerically.
2. Generate 1-year paths with the exact non-central chi-square transition.
3. Generate coarse monthly Euler paths from the same initial condition.
   At the first negative value, stop that path and record a failure.
   Do not take square roots of negative numbers.
4. Compare the Euler failure rate with the exact transition's non-negativity.
5. Shrink the step to check sensitivity.
6. Do not conclude Euler preserves positivity because no negatives appear
   in a sample. Explain also that using the same seed for different
   distributions does not by itself give the same Brownian path.
```

What to check

- Feller check
- Numerical failure rate
- Path charts
- Properties of the exact process vs the approximation

Shocks, mean reversion and discounting build rate models

Next week: apply the Kalman filter to Bank of Korea (BOK) GDP nowcasting

1

Brownian motion · SDE · Itô

Shocks grow with $\sqrt{\text{time}}$, and functions get an Itô correction term.

2

OU = exactly discretized AR(1)

Recover reversion speed, long-run mean and half-life from regressions — but check the conditions.

3

Bond price = average of path-wise discounts under Q

Vasicek and CIR give exponential-affine prices; report numerical and model errors separately.

GBM and OU return in Week 16 as classic generators for scenarios